MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, ULTRA-FAST RECOVERY,
HIGH VOLTAGE, POWER RECTIFIER,
IN6620 THROUGH IN6625 AND U SUFFIX VERSIONS,
JANTX, JANTXV, AND JANS

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a silicon, high voltage, ultra-fast recovery semiconductor rectifier diode, intended for use in modern, switched-mode power converters and motor control circuits. Three levels of product assurance are provided for each device as specified in MIL-S-19500.

1.2 Physical dimensions. See figure 1.

1.3 Maximum ratings. See columns 1 through 9.

1.4 Primary electrical characteristics. See columns 10 through 17. At T_A = +25°C unless otherwise noted.

Col 1	Col 2	Col 3	Col 4	Col 5	Col 6	Col 7	Col 8	Col 9
Device type	V_{RWM} V_R V_{RSM}	$I_{F(AV)1}$ at T_L = +55°C L = .375 inch (9.52 mm)	$I_{F(AV)2}$	I_{FSM} at t_p = 8.3 ms	Barometric pressure	t_{rr1} at 1/ I_F = 0.5 A I_R = 1.0 A I_{REC} = 0.25 A	T_{STG}	T_{OP}
	V pk V dc V pk	2/ 3/ 4/ 5/ A	2/ 6/ 7/ A	A pk	mm Hg	ns	°C	°C
1N6620 and U	200	2.0	1.2	20	8	30	-65 to +200	-65 to +175
1N6621 and U	400	2.0	1.2	20	8	30	"	"
1N6622 and U	600	2.0	1.2	20	8	30	"	"
1N6623 and U	800	1.5	1.0	20	33	50	"	"
1N6624 and U	900	1.5	1.0	20	33	50	"	"
1N6625 and U	1000	1.5	1.0	15	33	60	"	"

See footnotes at end of table.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Laboratory Command, Reliability, Logistics and Standardization Division, ATTN: SLCET-R-S, Fort Monmouth, NJ 07703-5000, by using the self-addressed Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Col 10	Col 11	Col 12	Col 13	Col 14	Col 15	Col 16	Col 17
Device type	I_{R1} at T_J = +25°C	I_{R2} at T_J = +150°C	$I_{RM(REC)}$ at 2 A, 100 A/µs	C_T at V_R = 10 V	$R_{\theta JL}$ at L = 0.375"	V_{FM1} at I_F = column 3	V_{FM2} at I_F = column 4
	µA	µA	A pk	pF	°C/W	V	V
1N6620 and U	0.5	150	3.5	10	38	1.60	1.40
1N6621 and U	0.5	150	3.5	10	38	1.60	1.40
1N6622 and U	0.5	150	3.5	10	38	1.60	1.40
1N6623 and U	0.5	150	4.2	10	38	1.80	1.55
1N6624 and U	0.5	150	4.2	10	38	1.80	1.55
1N6625 and U	1.0	200	5.0	10	38	1.95	1.75

1/ The reverse recovery time (method 3473, condition A, of MIL-STD-750) at T_J = +125°C shall not exceed three times the +25°C limit.

2/ Average current with a 50 percent duty cycle square wave including reverse voltage amplitude equal to the magnitude of full rated V_{RWM}.

3/ Derate linearly at 0.80 percent/°C for T_L greater than +55°C.

4/ Higher ratings are available with appropriate heat sinking or forced air cooling to keep the maximum junction temperature at or below +175°C as proven by the junction temperature rise test.

5/ These rated currents apply to U suffix types when the maximum temperature of the end caps (mounting surface) is +110°C; derate linearly at 1.5 percent/°C above +110°C.

6/ Derate linearly at 0.67 percent/°C for T_A greater than +25°C.

7/ This I_O rating is independent of heat sinking, special mounting, or forced air across the body or leads of the device.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARDS

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.
MIL-STD-1276 - Leads for Electronic Component Parts.

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

V_{FRM} - - - - - - - - - -	Forward recovery voltage
T_{CVF} - - - - - - - - - -	Temperature coefficient of forward voltage
$I_{RM(REC)}$ - - - - - - - - -	Peak reverse recovery current

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and in figure 1 herein. All devices shall be metallurgically bonded-thermally-matched-noncavity-double plug construction as defined in MIL-S-19500. U version shall be structurally identical to the non-U version except for lead configuration.

3.3.1 Lead finish. Lead finish shall be solderable as defined in MIL-S-19500 and herein. Where a choice of lead finish is desired, it shall be specified in the contract or purchase order (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

3.4.1 Marking of U version. For U version only, all marking (except see 3.5 below) may be omitted from the body, but shall be retained on the initial container.

3.5 Polarity. The polarity of all types shall be indicated with a contrasting color band to denote the cathode end. Alternatively, for U suffix devices, a minimum of three contrasting color dots spaced around the periphery on the cathode end may be used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500 and as specified herein.

4.2 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500.

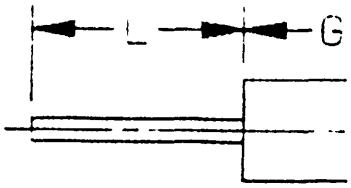
4.2.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. (End-point electrical measurements shall be in accordance with the applicable steps of table V herein.)

4.2.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500 and tables IIa and IIb herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table V herein.

4.2.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500 and table III herein. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table III herein.

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ontrolled.
and



Ltr	I_F
	Min
øB	.065
øD	.027
G	.125
L	.700

1 15	Col 16	Col 17
at 0.375"	V_{FM1} at I_F = column 3	V_{FM2} at I_F = column 4
C/W	V	V
38	1.60	1.40
38	1.60	1.40
38	1.60	1.40
38	1.80	1.55
38	1.80	1.55
38	1.95	1.75

750) at T_J = +125°C shall

ng reverse voltage amplitude

ced air cooling to keep the
junction temperature rise

perature of the end caps
bove +110°C.

or forced air across the

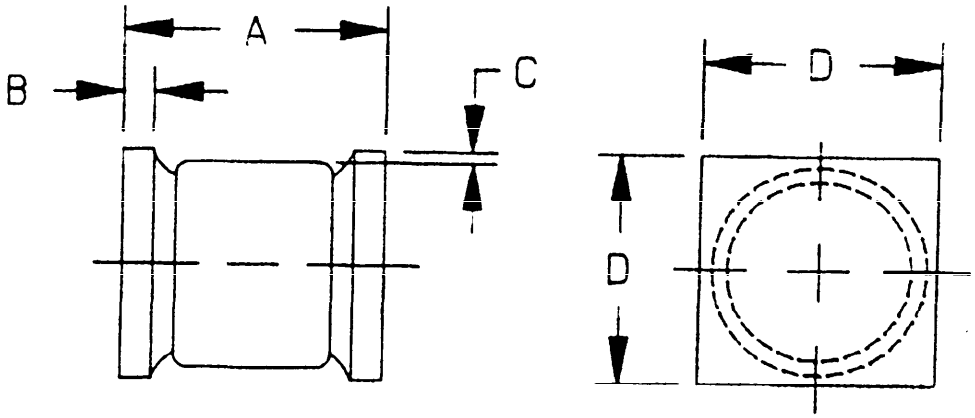
NOTES:
1. Metric equivalents are given for ge
2. Dimension G shall include the secti
   This uncontrolled area is defined a
   extending .050 (1.27 mm) onto the l
3. Dimension øB shall be measured at t

ifications, standards, and
Unless otherwise
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FIGURE 3

tion for.

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Dimensions				
Ltr	1N6620U through 1N6625U			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.168	.200	4.27	5.08
B	.019	.028	0.48	0.71
C	.003	---	0.08	---
D	.091	.103	2.31	2.62

FIGURE 1. Physical dimensions - Continued.

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4.3 Screening (JANTX, JANTXV, and JANS levels). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
9	I_{R1} and V_{FM1}	Not applicable
11	I_{R1} and V_{FM1}; ΔI_{R1} = 100 percent of initial reading or ±100 nA dc (±200 nA dc for 1N6625), whichever is greater. ΔV_{FM1} = ±0.05 dc	I_{RM1} and V_{FM1}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein: ΔI_{R1} = 100 percent of initial reading or ±100 nA dc (±200 nA dc for 1N6625), whichever is greater. ΔV_{FM1} = ±0.05 V dc. Scope display evalation (see 4.5.4).	Subgroup 2 of table I herein: I_{R1} = 100 percent of initial reading or ±100 nA dc (±200 nA dc for 1N6625), whichever is greater. ΔV_{FM1} = ±0.05 V dc. Scope display evaluation (see 4.5.4).

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: Method 1038, condition B, of MIL-STD-750, T_A = +25°C, V_{RWM} = rated (see 1.3); F = 60 Hz; I_O, the average of half-sine rectified (where nominal I_{Fpk} = 3.14 x I_O), = $I_{F(av)2}$ (see 1.3).

4.4 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500 and table IV, group E inspection, herein.

4.5 Methods of inspection. Methods of inspection shall be specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Inspection of conditions. Unless otherwise specified, all inspections shall be conducted at an ambient temperature T_A of +25°C ±3°C.

4.5.3 Life tests. These tests shall be conducted with a half-sine waveform of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall be neither greater than 180 degrees, nor less than 150 degrees.

4.5.3.1 Mounting conditions. The diodes shall be suspended by the leads with at least 6 inches of number 22 or smaller wire between each lead and the power source. The connecting wires may be soldered to the leads, or MS27491-20 contacts may be used. Other clips may be used provided they have equal or less mass than the MS27491-20 contact. No forced air shall be permitted and the diodes shall be shielded from drafts.

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4.5.3.2 Alternate mounting conditions. At the option of the manufacturer, other clips or heat sink mounting configurations may be utilized provided that I_O is adjusted such that the junction temperature of each diode is maintained at least +120°C above the specified ambient temperature of +25°C.

4.5.4 Scope display evaluation. The reverse breakdown characteristics shall be viewed on an oscilloscope with display calibration factors of 10 to 50 µA/division and 50 to 200 V/division. Reverse current over the knee shall be at least 50 µA and less than 500 µA peak. Any discontinuity below actual breakdown or dynamic instability of the trace shall be cause for rejection of that device.

4.5.5 Thermal resistance. Thermal resistance shall be measured in accordance with method 3101 of MIL-STD-750. The reference point shall be the lead temperature at .375 inch (9.52 mm) from the body of the device, and shall be held between +25°C and +85°C. The measuring current, IM shall be 1.00 mA dc for 1N6620 through 1N6625 and 4.00 mA for 1N6626 through 1N6631. Thermal resistance values shall be in accordance with 1.4.

4.5.6 Reverse recovery time.

4.5.6.1 Low current reverse recovery time. The low current reverse recovery time shall be measured in the circuit of figure 2 or equivalent. The recovery condition shall be 0.5 A forward current to 1.0 A reverse current. The reverse recovery time is defined as the time the rectifier begins to conduct in the reverse direction (crosses I = zero) until the reverse current decays to -0.25 A. The point of contact on the leads shall be no less than .375 inch (9.52 mm) from the diode body.

4.5.6.2 High current reverse recovery time. The high current reverse recovery time shall be measured in the circuit of figure 3 or equivalent. A pulse of forward current for the DUT is provided through S_1 and controlled by V_3, R_L and the input pulse V_1. The duration of pulse, t_1, should be sufficient to fully establish the stored charge associated with the specified I_F; it is suggested to be between 10 and 30 microseconds. The reverse voltage to the DUT is applied through S_2 at a rate determined by the rise time, t_2, and magnitude of the input pulse, V_2, which is applied slightly before the end of t_1. The negative supply, V_4, is to be specified. It should be sufficiently low to prevent the peak voltage, $V_{RM(REC)}$, from exceeding the rated reverse voltage, V_{RWM}, of the device under test. For most repeatable measurement, 30 V is recommended.

4.5.7 Junction temperature rise test. This test shall be performed in accordance with 4.5.7.1 and figure 4.

4.5.7.1 Test procedure. Determine the temperature coefficient of forward voltage (T_{CVF}) using a lower reference temperature (T_1) of +25°C to +30°C, an upper reference temperature (T_2) of +150°C to +200°C, and a reference current (I_{REF}) of 1 to 10 mA dc.

$$T_{CVF} = \frac{V_{F1} - V_{F2}}{T_2 - T_1}$$ Where $V_{F1} = V_F$ at I_{REF} at T_1 and $V_{F2} = V_F$ at I_{REF} at T_2.

At the option of the manufacturer, an average T_{CVF} based on a random sample of at least 25 production rectifiers may be used. To determine T_{CVF} with the diode mounting and temperature conditions specified, and with S_1 open ($I_O = 0$) read $V_{FO} = V_F$ at I_{REF}. Close S_1 and adjust the power source and R_L for the specified I_O. After thermal equilibrium has been established, read $V_{F3} = V_F$ at I_{REF} at 250 µs +50 µs after the power pulse. I_{REF} shall be within +1 percent of the value used for calibration. The rise in junction temperature is then computed as T_J rise $= \frac{V_{FO} - V_{F3}}{T_{CVF}}$.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits 3/		Unit
	Method	Conditions 2/		Min	Max	
Subgroup 1						
Visual and mechanical	2071					
Subgroup 2						
Forward voltage	4011	I_{FM1} = column 3 (peak) pulsed (see 4.5.1)	V_{FM1}		col 16	V
Forward voltage	4011	I_{FM2} = column 4 (peak) pulsed (see 4.5.1)	V_{FM2}		col 17	V
Reverse current leakage	4016	V_{RM} = column 2 (peak) pulsed (see 4.5.1)	I_{R1}		col 11	µA
Breakdown voltage	4021	I_R = 50 µA pulsed (see 4.5.1)	$V_{(BR)1}$	110% of col 2		V
Subgroup 3		T_A = +150°C				
High temperature operation						
Reverse current leakage	4016	V_{RM} = column 2 (peak) pulsed (see 4.5.1)	I_{R2}		col 12	µA
Low temperature operation		T_A = -65°C				
Forward voltage	4011	I_{FM3} = 50 percent of column 3 (peak) pulsed (see 4.5.1)	V_{FM3}		110% of col 16	V
Breakdown voltage	4021	I_R = 50 µA pulsed (see 4.5.1)	V_{BR2}	col 2		V

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits 3/		Unit
	Method	Conditions 2/		Min	Max	
Subgroup 4						
Forward recovery voltage	4026	I_F = 0.5 A t_r = 12 ns	V_{FRM}			V
1N6620 and U					12	
1N6621 and U					12	
1N6622 and U					12	
1N6623 and U					18	
1N6624 and U					18	
1N6625 and U					30	
Capacitance	4001	V_R = 10 V f = 0.1 to 1 MHz	C_T			pF
1N6620 and U					10	
1N6621 and U					10	
1N6622 and U					10	
1N6623 and U					10	
1N6624 and U					10	
1N6625 and U					10	
Reverse recovery time	4031	See 4.5.6.1, condition B	t_{rr1}			ns
1N6620 and U					30	
1N6621 and U					30	
1N6622 and U					30	
1N6623 and U					50	
1N6624 and U					50	
1N6625 and U					60	
Reverse recovery time	4031	See 4.5.6.2, I_F = 2 A di/dt = 100 A/µs	t_{rr2}			ns
1N6620 and U					45	
1N6621 and U					45	
1N6622 and U					45	
1N6623 and U					60	
1N6624 and U					60	
1N6625 and U					90	

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits 3/		Unit
	Method	Conditions 2/		Min	Max	
Subgroup 4 - Continued						
Peak recovery current		See figure 3	$I_{RM(REC)}$			A
1N6620 and U					3.5	
1N6621 and U					3.5	
1N6622 and U					3.5	
1N6623 and U					4.2	
1N6624 and U					4.2	
1N6625 and U					5.0	
Subgroups 5, 6, and 7						
Not applicable						

1/ For sampling plan, see MIL-S-19500.

2/ See 1.3, columns 1 through 9.

3/ See 1.4, columns 10 through 17.

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TABLE IIa. Group B inspection for JANS devices.

Inspection 1/	MIL-STD-750	
	Method	Conditions 2/ 3/
Subgroup 1		
Physical dimensions	2066	See figure 1
Subgroup 2		
Solderability	2026	
Resistance to solvents	1022	
Subgroup 3		
Thermal shock (glass strain)	1056	Condition A, T_{low} = 0°C, T_{high} = +100°C, 10 cycles
Surge current	4066	I_{FSM} = rated (see 1.3); 10 surges of 8.3 ms each at 1 minute intervals
		I_O = 0, V_{RSM} = 0
Hermetic seal Gross leak	1071	
Electrical measurements		See table V, steps 1 and 2
Subgroup 4		
Intermittent operation life	1037	T_A = +25°C, I_O = 4.3.1 V_{RM} = column 2 t_{on} = t_{off} = 3 minutes minimum, 2000 cycles
Electrical measurements		See table V, steps 1, 2, 3, and 6
Subgroup 5		
Accelerated steady-state operation life	1027	T_A = +25°C, no heat sink, I_F = 1.5 x column 4 minimum; adjust T_A or I_F as required to achieve T_J = +275°C, +0, -35°C, t = 96 hours
Electrical measurements		See table V, steps 1, 4, 5, and 6

See footnotes at end of table.

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TABLE IIa. Group B inspection for JANS devices - Continued.

Inspection 1/	MIL-STD-750	
	Method	Conditions 2/ 3/
Subgroup 6		
Thermal resistance	3101	See 4.5.5 $R_{\theta JL}$ = column 15 of 1.4
Junction temperature rise		See 4.5.7

1/ For sampling plan, see MIL-S-19500.

2/ See 1.3, columns 1 through 9.

3/ See 1.4, columns 10 through 17.

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TABLE IIb. Group B inspection for JANTX and JANTXV devices.

Inspection 1/	MIL-STD-750	
	Method	Conditions 2/
Subgroup 1		
Solderability	2026	
Resistance to solvents	1022	
Subgroup 2		
Thermal shock (temperature cycling)	1051	45 cycling including screen
Surge current	4066	I_{FSM} = rated (see 1.3); 10 surges of 8.3 ms each at 1 minute intervals
		I_O = 0, V_{RSM} = 0
Hermetic seal Gross leak	1071	
Electrical measurements		See table V, steps 1, 2, and 3
Subgroup 3		
Steady-state operation life	1027	T_A = +25°C, I_O = 4.3.1, V_{RM} = column 2 (see 1.3), t = 340 hours
Electrical measurements		See table V, steps 1 and 2
Subgroup 4		
Not applicable		
Subgroup 5		
Not applicable		
Subgroup 6		
High-temperature life (nonoperating)	1032	T_A = +200°C, t = 340 hours
Electrical measurements		See table V, steps 1, 2, and 6

1/ For sampling plan, see MIL-S-19500.

2/ See 1.3, columns 1 through 9.

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TABLE III. Group C inspection for JANTXV devices.

Inspection 1/	MIL-STD-750	
	Method	Conditions 2/
Subgroup 1		
Physical dimensions	2066	See figure 1
Subgroup 2		
Thermal shock (glass strain)	1056	Condition A
Terminal strength (tension) 3/	2036	Test condition A; weight = 12 pounds
Hermetic seal Gross leak	1071	
Moisture resistance	1021	
Electrical measurements		See table V, steps 1 and 2
Subgroup 3		
Shock	2016	
Vibration, variable frequency	2056	Nonoperating
Electrical measurements		See table V, steps 1 and 2
Subgroup 4		
Salt atmosphere (corrosion)	1041	
Subgroup 5		
Not applicable		
Subgroup 6		
Steady-state operation life	1026	T_A = +25°C, I_O = 4.3.1 V_{RWM} = column 2
Electrical measurements		See table V, steps 1, 2, 3, and 6

1/ For sampling plan, see MIL-S-19500.

2/ See 1.3, columns 1 through 9.

3/ Not applicable to U suffix types.

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TABLE IV. Group E inspection for (all quality levels) for qualification only.

Inspection	MIL-STD-750		Sampling plan
	Method	Conditions	
Subgroup 1			22 devices, c = 0
Thermal shock (glass strain)	1056	2000 cycles	
Hermetic seal Gross leak	1071		
Electrical measurements		See table V, steps 1 and 2	
Subgroup 2			22 devices, c = 0
Blocking life	1038	80 to 85 percent of rated V_R, T_A = +150°C, t = 1000 hours +65, -0 hours	
Subgroup 3			3 devices, c = 0
DPA		Photos of cross sections shall be submitted in the qualification report. Vendor shall retain duplicate photos. The photo shall be of sufficient size to provide a 1.5 inch (38.10 mm) minimum view of the entire die to plug interfaces. The section may be stained to enhance the junction area and die to plug interface.	
Subgroup 4 1/			22 devices, c = 0
Thermal resistance, junction to lead	3101	L = .375 inch (9.52 mm) Limits: 38°C/W for 1N6620 through 1N6625	
Subgroup 5			22 devices, c = 0
Barometric pressure (reduced)	1001	1N6620 through 1N6622 at 8 mm Hg. 1N6623 through 1N6625 at 33 mm Hg.	
Subgroup 6			
Not applicable			
Subgroup 7			22 devices, c = 0
Junction temperature rise test		See 4.5.7	

1/ For U suffix types thermal resistance, junction to end caps.

Limits:
20°C/W for 1N6620U through 1N6625U

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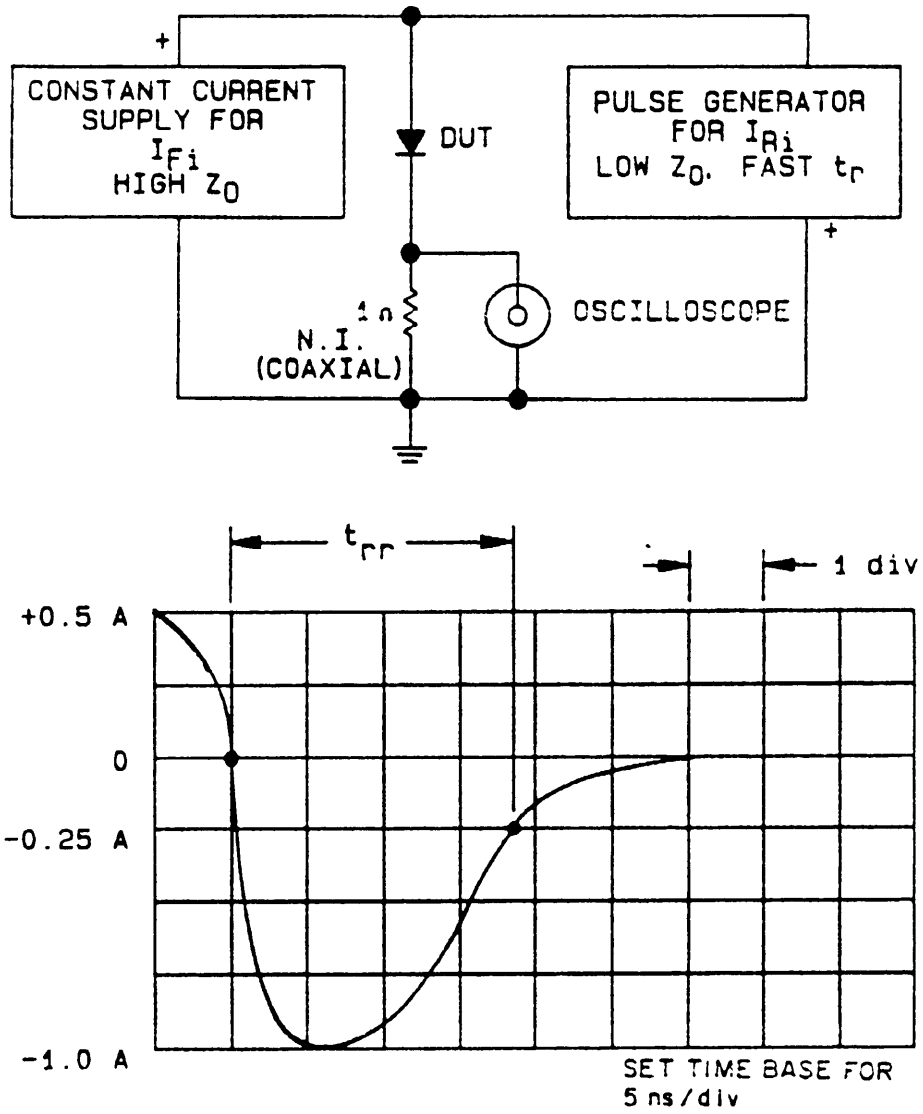
TABLE V. Groups A, B, C, and E electrical measurements.

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions 1/		Min	Max 1/ 2/	
1.	Forward voltage	4011	I_{FM} = column 3 pulsed (see 4.5.1)	V_{FM1}		col 16	V
2.	Reverse current leakage	4016	V_{RM} = column 2	I_{RM1}		col 11	µA
3.	Forward voltage	4011	I_{FM} = column 4 pulsed (see 4.5.1)	V_{FM2}		col 17	V
4.	Forward voltage change	4011	I_F = column 4 pulsed (see 4.5.1)	ΔV_F	Change from previous measured value	±50	mV
5.	Reverse current change	4016	V_{RM} = column 2	ΔI_R			
	1N6620 through 1N6625					+1	µA
6.	Reverse recovery time	4031	See 4.5.6.1	t_{rr1}		col 7	ns

1/ See 1.3, columns 1 through 9.

2/ See 1.4, columns 10 through 17.

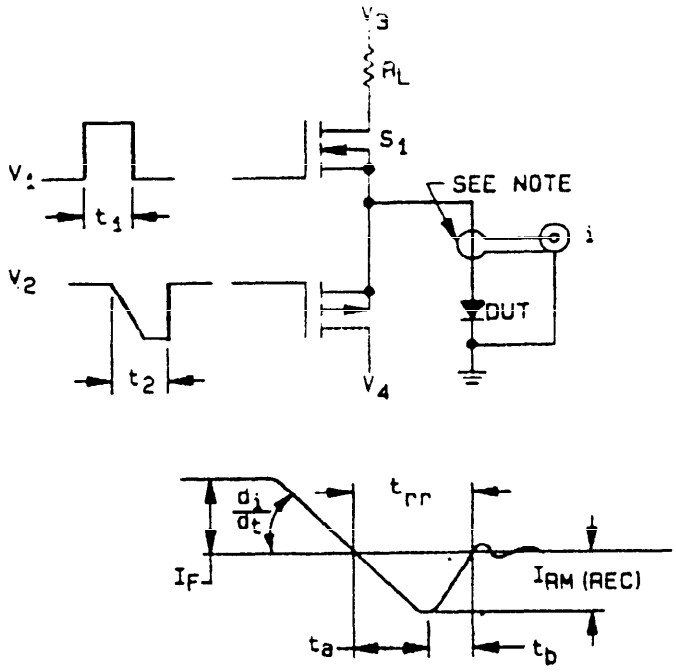
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NOTES:
1. Oscilloscope: Rise time ≤ 3 ns; input impedance = 1 MΩ; 22 pF.
2. Pulse generator: Rise time ≤ 8 ns; source impedance = 10Ω.

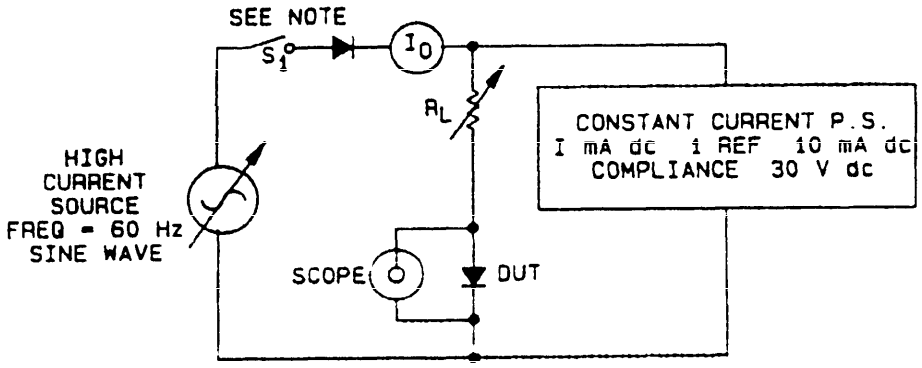
FIGURE 2. Reverse recovery time test circuit and characteristic waveform.

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NOTE: Current sensing transformer; alternatively a low inductance resistor may be used, from cathode to ground, for current versus time measurements.

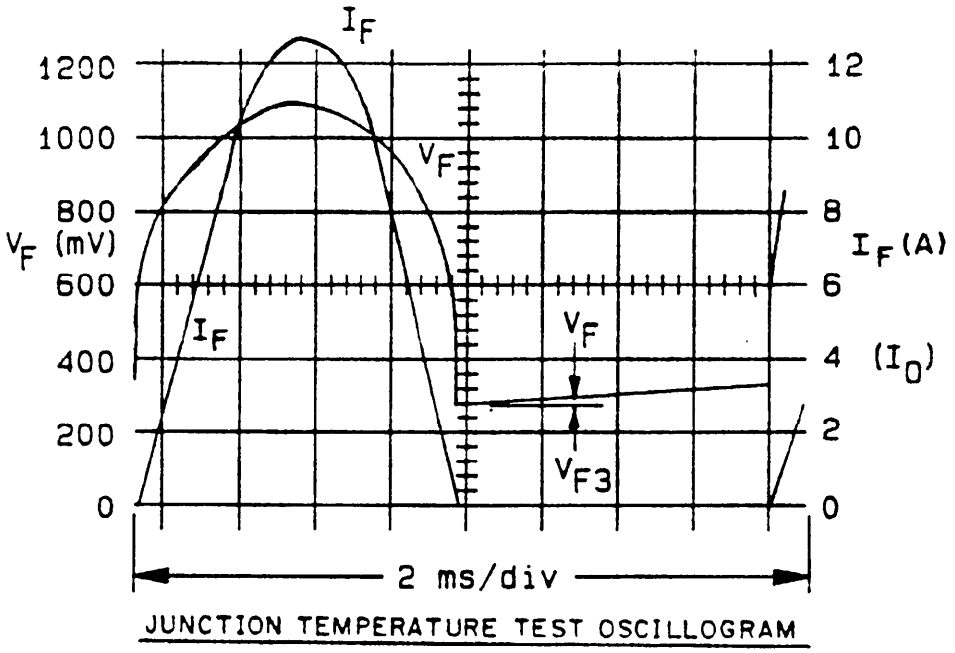
FIGURE 3. High current reverse recovery test circuit and waveforms.



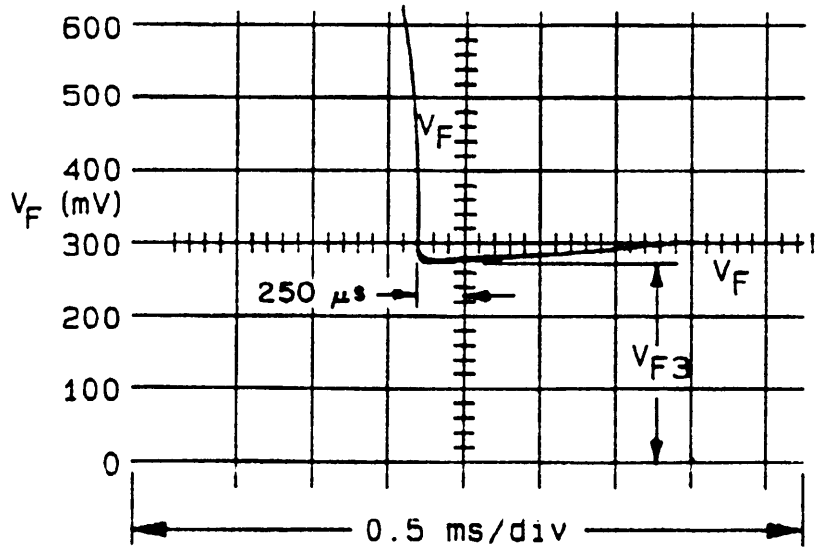
NOTE: Blocking diode shall have a forward current rating $\geq$ 2 x I_{O1}.

FIGURE 4. Junction temperature rise test circuit and characteristic waveforms.

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JUNCTION TEMPERATURE TEST OSCILLOGRAM



EXPANDED OSCILLOGRAM OF V_F

FIGURE 4. Junction temperature rise test circuit and characteristic waveforms - Continued.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Lead finish may be specified (see 3.3.1).

6.3 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's part number. This information in no way implies that manufacturer's part numbers are suitable as a substitute for the military part or identifying number (PIN).

PIN	Manufacturer's CAGE code	Manufacturer's and user's PIN

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Air Force - 11, 19, 85
DLA - ES
NASA - MSFC - EG02

User activities:
Army - SM
Navy -AS, CG, MC, OS
Air Force - 13
NASA - NA

Preparing activity:
Army - ER

Agent:
DLA - ES

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